

EXPLORATIONS IN THE F-SIGNATURE AND LIMIT F-SIGNATURE OF HOMOGENIZED ORTHOGONAL POLYNOMIALS

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ABSTRACT. We study two numerical invariants related to how we quantify singularities of polynomials: the F-signature and limit F-signature. Each of these encodes information about the F-pure threshold and log canonical threshold. Motivated by the appearance of the Hermitian polynomial in the quantum harmonic oscillator, we elected to explore a broad class of orthogonal polynomials: homogenized Hermitian, Laguerre, and Chebyshev polynomials. Using the Macaulay 2 programming language, we made several key approximations that led to the discovery of a piecewise quadratic function. This was not a coincidence: a theorem already existed. The result depends on the number of distinct linear factors a polynomial has. However, as we will see, this breaks when changing how we homogenize.

1. INTRODUCTION

In the fields of algebraic geometry and commutative algebra, there is great importance in the concept of singularities. Suppose we have a polynomial $f \in \mathbb{Z}[x_1, \dots, x_n]$. If all first order partial derivatives $\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}$ vanish at a point p , then p is a singularity of f . Singularities are where the geometry of $f = 0$ breaks: cusps, intersections, and sharp corners. When a point is not singular, it may also be called "smooth". One may then ask: how are singularities quantified? Using number theory, we can make the task of answering this question much easier by reducing the coefficients of f modulo p . With our new tool, this question leads us to the F-pure threshold and the log canonical threshold (see Definitions 2.11 and 2.12).

In recent discoveries, mathematicians have identified two numerical invariants: the F-signature and the limit F-signature. Both of these encode specific information about the log canonical threshold and F-pure threshold. A pressing question now is what these numerical invariants are equal to for specific polynomials or even families of polynomials. Hermitian polynomials are of interest due to their usage in quantum mechanics, which led us to explore other orthogonal polynomials with physical applications. When calculating the limit F-signature for homogenized Hermitian, Laguerre, and Chebyshev polynomials, the limit F-signature functions follows a piecewise second degree polynomial. Lastly, the evaluations of the F-signature and limit F-signature depend on how the polynomial is homogenized.

ACKNOWLEDGMENTS

We are thankful to have been supported by our instructor Anna Brosowsky, who taught us several key concepts, answered questions, and provided feedback on our hypotheses, methods, and final organization.

The University of Nebraska-Lincoln’s Holland Computing Center enabled collection of data generated by Macaulay2 code through permission to use the Swan cluster.

AI Disclosure. The computations that led to our conjectures and formulations were a result of code supplemented by the use of Claude Sonnet 5, namely in the scaffolding of Macaulay2 code. GPT-5.5 also contributed to the creation and debugging of code for a custom data aggregation tool.

2. BACKGROUND

2.1. Hermitian Polynomials. We elected to explore the Hermitian family of polynomials because they appear in the set of solutions to the time-independent Schrödinger equation for the quantum harmonic oscillator. That is, the Schrödinger equation [RL19] (1) with a the Hamiltonian operator [So26] ($\hat{H}$) in equation (2).

$$(1) \quad \hat{H}|\psi(\hat{x})\rangle = E|\psi(\hat{x})\rangle$$

$$(2) \quad \hat{H} = \frac{\hat{p}_x^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

where $\hat{p}_x$ is the x -direction momentum operator $-\frac{i\hbar\partial}{\partial x}$, ω is the oscillator angular frequency, m is the particle mass, and $\hat{x}$ is the position of the particle. The wave equations that satisfy this equation are of the form

$$\psi(x) = H_n(x)e^{-\frac{m\omega x^2}{\hbar}} \cdot \sqrt{\frac{m\omega}{\pi\hbar}} \frac{1}{\sqrt{2^n n!}}$$

where n is the principal quantum number and $\hbar$ is the adjusted Planck constant. For the purposes of this paper, we will consider the Hermitian polynomial in isolation from the other terms in this equation.

An additional case that will be explored is the three-dimensional harmonic oscillator¹, with solutions of the form

$$\psi(x, y, z) = X(x, n_x) \cdot Y(y, n_y) \cdot Z(z, n_z)$$

where each component is the one-dimensional wave equation in their respective dimension, and each n_j is the principal quantum number in the direction of its associated component. This wave function satisfies the Schrödinger equation for the Hamiltonian²

$$\hat{H} = \frac{\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2}{2m} + \frac{1}{2}m\omega^2(\hat{x}^2 + \hat{y}^2 + \hat{z}^2)$$

The three-dimensional wave function described contains a term of

$$H_\psi := H_{n_x, n_y, n_z}(x, y, z) = H_{n_x}(x)H_{n_y}(y)H_{n_z}(z)$$

¹Particularly, the Cartesian three-dimensional harmonic oscillator. Although there are radial and cylindrical forms, the Hermitian polynomials are relevantly applicable to the Cartesian case.

²Note that ω does not change with direction, by assumption. This means that the harmonic oscillator strength will be equal on each direction, and therefore symmetric.

which we explored in isolation from the exponential and constant terms due to the symmetry given by the chosen Hamiltonian.

Hermitian polynomials belong to a broader field of *orthogonal* polynomials. Two polynomials f and g if $\langle f, g \rangle = 0$ for some inner product. Each orthogonal polynomial has an orthogonality measure of the form

$$d(\mu) = cw(x)dx$$

where c is a constant defined by

$$c = \left(\int_R w(x)dx \right)^{-1}$$

and $w(x)$ is the weight function. For Hermitian polynomials specifically, the weight function is $w(x) = e^{-x^2}$ while $c = \Gamma(\frac{1}{2}) = \pi^{-\frac{1}{2}}$. [DX14]

Definition 2.1. For $n \geq 0$, the n -th Hermitian polynomial is defined as

$$(3) \quad H_n(x) = (-1)^n e^{x^2} \left(\frac{d^n}{dx^n} e^{-x^2} \right).$$

We may then recursively define Hermitian polynomials via

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x).$$

2.2. Other Orthogonal Polynomials and Homogenization.

Definition 2.2. ([DX14]) The *Chebyshev polynomials* are a set of orthogonal polynomials defined recursively via

$$\begin{aligned} T_0(x) &= 1 \\ T_1(x) &= x \\ T_{n+1}(x) &= 2xT_n(x) - T_{n-1}(x). \end{aligned}$$

Definition 2.3. ([DX14]) The *Laguerre polynomials* are a set of orthogonal polynomials defined recursively via

$$\begin{aligned} L_0(x) &= 1 \\ L_1(x) &= 1 - x \\ L_{k+1}(x) &= \frac{(2k+1-x)L_k(x) - kL_{k-1}(x)}{k+1}. \end{aligned}$$

These typically have fractional coefficients, but these simply act as units in $\mathbb{F}_p$.

When calculating the F-signature and limit F-signature, we must use polynomials that are members of $\mathfrak{m} = \langle x_1, \dots, x_n \rangle$. Thus, they must be homogeneous.

Definition 2.4. *Single-variable homogenization.* Given a polynomial $f(x_1, \dots, x_n) \in K[x_1, \dots, x_n]$, f may be *homogenized* with respect to y by

$$(4) \quad f^h = y^d f \left(\frac{x_1}{y}, \dots, \frac{x_n}{y} \right) \text{ where } d := \max\{\deg_{x_1} f, \dots, \deg_{x_n} f\}$$

Definition 2.5. *Multi-variable homogenization.* Given a polynomial $f(x_1, \dots, x_n) \in K[x_1, \dots, x_n]$, f may be *homogenized* with respect to $w_1, \dots, w_n$ by

$$(5) \quad f^h = w_1^{a_1} \cdots w_n^{a_n} f \left(\frac{x_1}{w_1}, \dots, \frac{x_n}{w_n} \right) \text{ where } a_i := \deg_{x_i} f$$

$$= \max \left\{ c_i | \exists b_1, \dots, b_i, \dots, b_n \text{ such that } x_1^{b_1} \cdots x_{i-1}^{b_{i-1}} \cdot x_i^{c_i} \cdot x_{i+1}^{b_{i+1}} \cdots x_n^{b_n} \text{ is a term of } f \right\}$$

2.3. Further Definitions.

Definition 2.6 ([BST13]). Let $f \in \mathbb{F}_p[x_1, \dots, x_n]$ be a homogeneous polynomial contained in $\mathfrak{m} = \langle x_1, \dots, x_n \rangle$. Then the *F-signature function* is defined for $t = \frac{k}{p^e}$ as

$$\mathfrak{s}_f\left(\frac{k}{p^e}\right) = 1 - \frac{1}{p^{en}} \dim_{\mathbb{F}_p} \left(\frac{\mathbb{F}_p[x_1, \dots, x_n]}{\langle x_1^{p^e}, \dots, x_n^{p^e}, f^k \rangle} \right).$$

For general $t \geq 0$, the formula is

$$\mathfrak{s}_f(t) = \lim_{e \rightarrow \infty} \mathfrak{s}_f\left(\frac{\lfloor tp^e \rfloor}{p^e}\right).$$

Definition 2.7 ([BST13]). Let $f \in \mathbb{Z}[x_1, \dots, x_n]$ be a homogeneous polynomial contained in $\mathfrak{m} = \langle x_1, \dots, x_n \rangle$. Then the *limit F-signature function* of f at $t \in \mathbb{R}_{\geq 0}$ is, if it exists,

$$\mathfrak{s}_f(t) = \lim_{p \rightarrow \infty} \mathfrak{s}_{f \bmod p}(t),$$

where $f \bmod p$ is f when viewed in $\mathbb{F}_p[x_1, \dots, x_n]$ after taking all the coefficients modulo p .

Lemma 2.8 ([BCPT26, Lemma 2.5]). *Let $f \in \mathbb{Z}[x_1, \dots, x_n]$ be a homogeneous polynomial, $f \neq 0$ and $f \neq 1$. Let $\Phi_f(t)$ be a continuous, non-increasing function on $[0, 1]$ with $\Phi_f(0) = 1$ and $\Phi_f(1) = 0$. If*

$$1 - \mathfrak{s}_{f \bmod p}\left(\frac{\lfloor tp \rfloor}{p}\right) = \Phi_f(t)$$

for all $t \in \mathbb{Q}_{\geq 0}$, then the limit F-signature function for f exists and is equal to $\Phi_f(t)$.

Lemma 2.9 ([BCPT26, Lemma 2.5]). *Let $f \in R = \mathbb{Z}[x_1, \dots, x_n]$ with $0 \neq f \in \mathfrak{m} = \langle x_1, \dots, x_n \rangle$. Let $\Psi(t)$ a continuous, non-increasing function on $[0, 1]$ with $\Psi(0) = 1$ and $\Psi(1) = 0$, such that for each $t \in \mathbb{Q}_{\geq 0}$, there is sequence $r_p(t)$ of non-negative real numbers such that:*

- $r_p(t) \leq t$ for each $p \gg 0$.
- $r_p(t) \rightarrow t$ as $p \rightarrow \infty$.

If $\lim_{p \rightarrow \infty} \mathfrak{s}_{f \bmod p}(r_p(t)) = \Psi(t)$ for all $t \in \mathbb{Q}_{\geq 0}$, then the limit F-signature function for f exists and is equal to $\Psi(t)$.

Theorem 2.10 ([BST13, Thm. A,B]). *Let f be a homogeneous polynomial in the ring $\mathbb{F}_p[x_1, \dots, x_n]$. Then the F-signature function $\mathfrak{s}_f(t)$ is a continuous and concave up function on $[0, \infty)$.*

Definition 2.11. Let f be a polynomial in $\mathbb{F}_p[x_1, \dots, x_n]$. Then the *F-pure threshold of f (at the origin)* is the rational number

$$\text{fpt}(f) = \sup \left\{ \frac{k}{p^e} \mid f^k \notin \langle x_1^{p^e}, \dots, x_n^{p^e} \rangle \right\} = \inf \left\{ \frac{k}{p^e} \mid f^k \in \langle x_1^{p^e}, \dots, x_n^{p^e} \rangle \right\}.$$

Definition 2.12. Let f be a polynomial in $\mathbb{C}[x_1, \dots, x_n]$. Then the *log canonical threshold f (at the origin)* is the number

$$\text{lct}(f) = \sup \left\{ \lambda \in \mathbb{R}_{>0} \mid \exists \text{ open ball } B \text{ around the origin s.t. } \int_B \frac{1}{|f|^{2\lambda}} < \infty \right\}.$$

Theorem 2.13 ([MTW05, Thm. 3.3,3.4]). *Let f be a polynomial in $\mathbb{Z}[x_1, \dots, x_n]$. Then*

$$\text{lct}(f) = \lim_{p \rightarrow \infty} \text{fpt}(f \bmod p).$$

Further, for all $p \gg 0$ we have $\text{fpt}(f \bmod p) \leq \text{lct}(f)$.

3. DATA AND FINDINGS

3.1. Computations of the Limit F-signature of Homogenized Orthogonal Polynomials. Initial computations tested the limit F-signature of single-variable homogenized Hermitian, Chebyshev, and Laguerre polynomials for $t = \frac{1}{d}$ with $d = 1, 2, \dots, 11$. During computations, we approximated the limit F-signature by letting $p = 997$ and $e = 1$. Ideally we would use a far higher e value, but computational limits made this impossible for our chosen prime, and it would not have changed our data in a meaningful way. After analyzing our first batch of data for $d = 1, 2, \dots, 11$ and $n = 1, 2, \dots, 11$, we used a custom data aggregation tool to make our results easier to work with. After plugging our data into Excel, our first correlation was discovered (see Figure 1). After discovering this correlation, we arbitrarily chose the 9th homogenized Hermitian polynomial and used MATLAB to remove the points where the limit F-signature was approximated to be 0. We then discovered a new correlation: the "black magic polynomial". Our initial conjecture was as follows: Let n be the degree of a homogenized orthogonal polynomial. Then we describe the limit F-signature:

$$(6) \quad \mathfrak{s}_f(t) = \begin{cases} (\frac{nt}{2} - 1)^2 & \text{for some small t-value} \\ 0 & \text{for some large t-value} \end{cases}$$

This function fulfills the requirements of our theorems and lemmas. Namely, it is concave up and continuous.

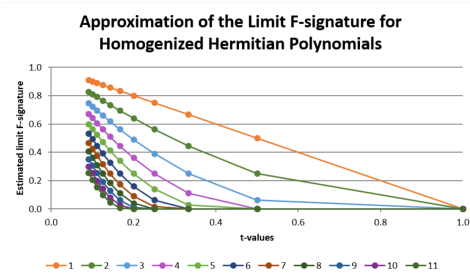


FIGURE 1. Limit F-signature of the first 11 homogenized Hermitian polynomials at $t = 1, \frac{1}{2}, \dots, \frac{1}{11}$. Approximated by $p = 997, e = 1$.

After the discovery of this piecewise polynomial, we calculated an approximation of the limit F-signature for homogenized Chebyshev and Laguerre polynomials, which also followed the equation (see Figures 2 and 3). We immediately began looking for literature that may already contain a proof for the polynomial, leading us to the following theorem proved by Eric Canton:

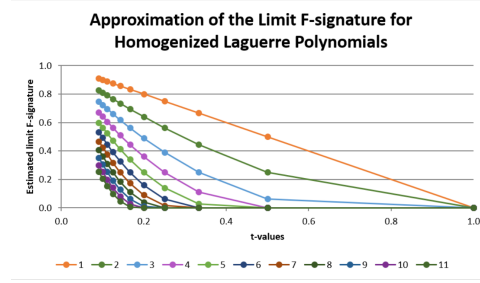


FIGURE 2. Limit F-signature of the first 11 homogenized Laguerre polynomials at $t = 1, \frac{1}{2}, \dots, \frac{1}{11}$. Approximated by $p = 997, e = 1$

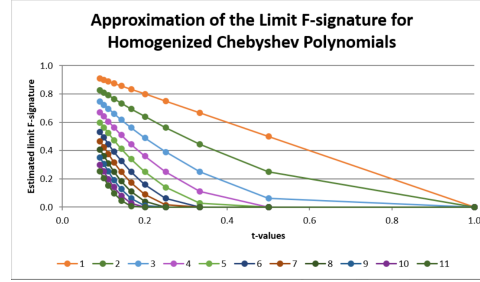


FIGURE 3. Limit F-signature of the first 11 homogenized Chebyshev polynomials at $t = 1, \frac{1}{2}, \dots, \frac{1}{11}$. Approximated by $p = 997, e = 1$

Theorem 3.1. [Can12] *The F-signature of the pair $s(R, f^t)$ where f is the product of at least two distinct linear factors converges uniformly on the interval $[0, c]$ (where c is the F-pure threshold) to the polynomial $\frac{n^2}{4}t^2 - nt + 1$ as $p \rightarrow \infty$.*

We choose to omit the proof due to advanced terminology from the field of commutative algebra. Thus, the relation of the limit F-signature between homogenized Hermitian, Laguerre, and Chebyshev polynomials follow from the number of their distinct linear factors. As a side note, the theorem uses the F-pure threshold as it is written, but in some sense we could use the log canonical threshold. Generally, there is some confusion due to how specific primes allow us to factor the polynomial. From here, we attempted to expand our understanding of this theorem by testing it in as many cases as possible, including the multivariate and nonhomogeneous cases.

3.2. F-Signature of Homogenized Two-Variable Hermitian Polynomials.

First, consider the multivariate cases. For our purposes, we remain only considering the Hermitian polynomials. For the two variable case, the distribution still followed Canton's polynomial. We first defined a two variable case of the Hermitian polynomials recursively:

$$\begin{aligned} H_0(x, y) &= 1 \\ H_1(x, y) &= 2x \\ H_n(x, y) &= 2xH_{n-1}(x, y) - 2y^2(n-1)H_{n-2}(x, y). \end{aligned}$$

Below is a two variable case of the 9th Hermite, showing that it follows Canton's theorem as it maps to $(\frac{9}{2}x - 1)^2$.

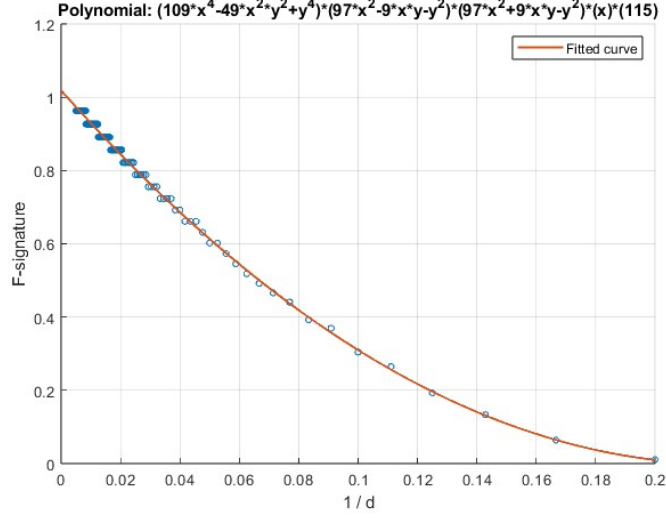


FIGURE 4. Graph of an approximation of the limit F-signature of a homogenized two variable case of the 9th Hermitian Polynomial, and the fitted curve $f(x) = 20.42x^2 - 9.124x + 1.018$.

3.3. F-Signature of Homogenized Three-Variable Hermitian Polynomials. Further, considering a three variable case of the Hermitian polynomials, Canton's polynomial notably does not properly fit the function. We defined this case recursively:

$$\begin{aligned} H_0(x, y, z) &= 1 \\ H_1(x, y, z) &= 2x \\ H_n(x, y, z) &= 2xH_{n-1}(x) - 2(y^2 + z^2)(n-1)H_{n-2}(x, y, z). \end{aligned}$$

The following is the 9th Hermitian polynomial, with the line of best fit as $13.93x^2 - 7.713x + 1.045$ instead of the predicted $(\frac{9}{2}x - 1)^2$.

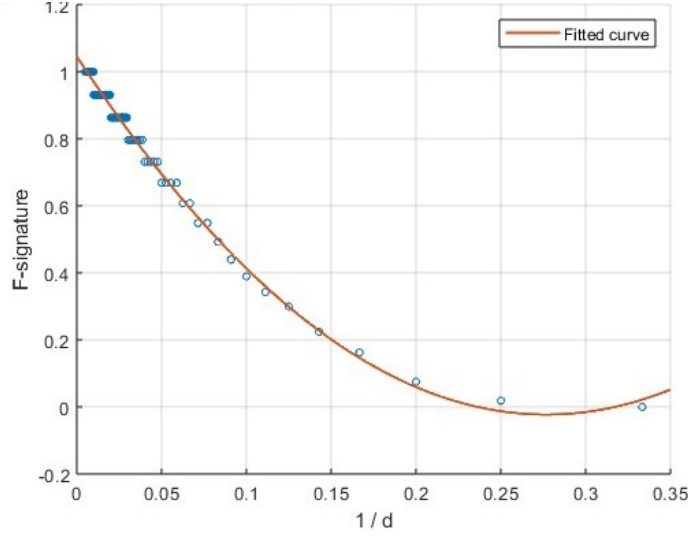


FIGURE 5. Graph of an approximation of the limit F-signature of a homogenized three variable case of the 9th Hermitian Polynomial.

3.4. F-Signature of the Product of Independently-Homogenized Hermitian Polynomials. In connection with the symmetric Cartesian 3-Dimensional quantum harmonic oscillator, we considered the case of H_ψ , with each Hermitian independently homogenized with respect to its own unique homogenizing variable. In all, this gave a homogeneous polynomial in 6 variables.

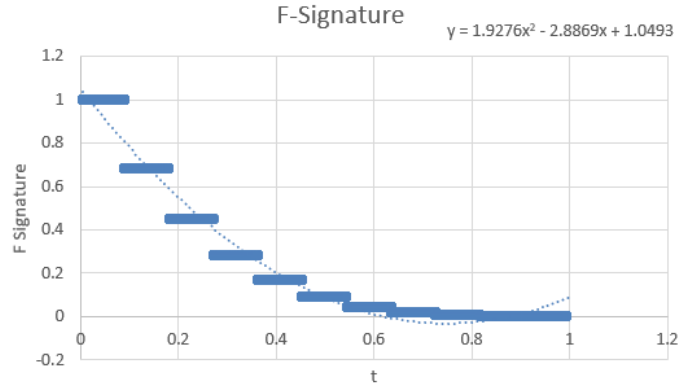


FIGURE 6. Graph of an approximation of the limit F-signature of H_ψ with $n_x = 1, n_y = 1, n_z = 2$ and the zeroes removed.

The data here was perfectly reproduced in the case of $n_x = 1, n_y = 2$, and $n_z = 1$, giving degeneracy in the F-signature for the same set of values for each n_i . This degeneracy was also observed to hold for other values of n_i . Our expectation was that this degeneracy would be present, since we assumed our harmonic oscillator to

be symmetric across all three directions, and the energy eigenstates for a particle with a given three n_i are degenerate in the same way under the assumed oscillator. We conjecture that this is due to the Hermitians' dependence on distinct linear factors.

The trend in Figure 6 is vaguely quadratic in nature, however further study would be required to confirm this. Artifacts from the floor function approximation are especially strong, since the Macaulay2 code generating the data did not account for duplicate data points. This created a skew toward zero which could not be rectified due to the prohibitive size of the data set coupled with time restrictions.

3.5. F-Signature of Homogenized Three-Variable Hermitian Products.

Another case in which three variables are considered is the product of multiple Hermitian polynomials. We split these into two different cases we tested. In both cases, we define 3 different Hermitian polynomials defined by variables x, y, z . For one case, we homogenize all functions by a fourth variable w , then take the product (see Section 3.6). We will consider a different case, where we homogenize two of the polynomials by x , then keep the third polynomial non-homogenized. This case turns out to be not symmetric, meaning the limit F-signature is independent of the sum of the degrees. However, when highlighting by the lowest degree, we see the data is the same as the case in Section 3.6 (refer to Figures 7, 8, and 9).

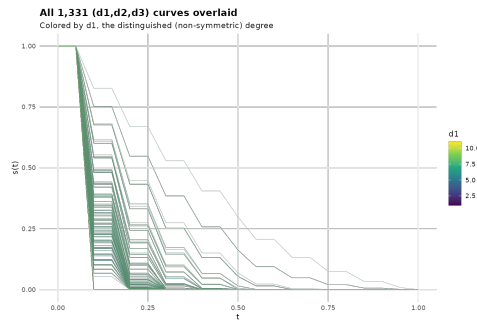


FIGURE 7. Limit F-signature of the product of two homogenized Hermitian polynomials with a non-homogeneous Hermitian polynomial, highlighted by the non-homogeneous degree. Made with R.

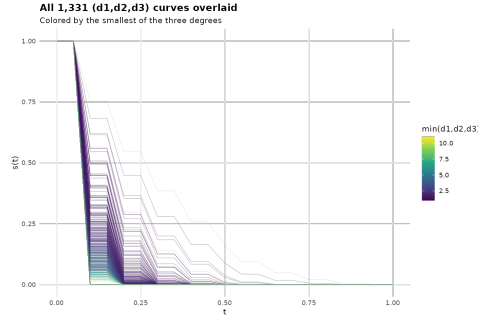


FIGURE 8. Limit F-signature of the product of two homogenized Hermitian polynomials with a non-homogeneous Hermitian polynomial, highlighted by lowest degree of the three. Made with R.

Thus, when analyzing the data, it is clear this case breaks the black magic polynomial.

3.6. F-Signature of Partially Homogenized Four-Variable Hermitian Products. As discussed in the previous section, we are dealing with a symmetric case. We define three homogenized Hermitian polynomials by x, y , and z respectively, homogenizing with a fourth variable w . We then take the limit F-signature of the product, giving us yet another result that breaks the black magic polynomial (see Figure 9).

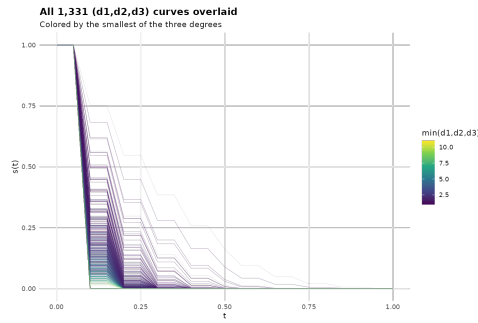


FIGURE 9. Limit F-signature of the product of three homogenized Hermitian polynomials defined by x, y , and z respectively, homogenized by a fourth variable w . Highlighted by lowest degree of the three. Made with R.

3.7. F-Signature of Nonhomogeneous Hermitian Polynomials. Next, consider the variation that can arise when the polynomial is nonhomogeneous. Interestingly, when the polynomial has an even number of linear factors, the limit F-Signature is 1 for all t values.

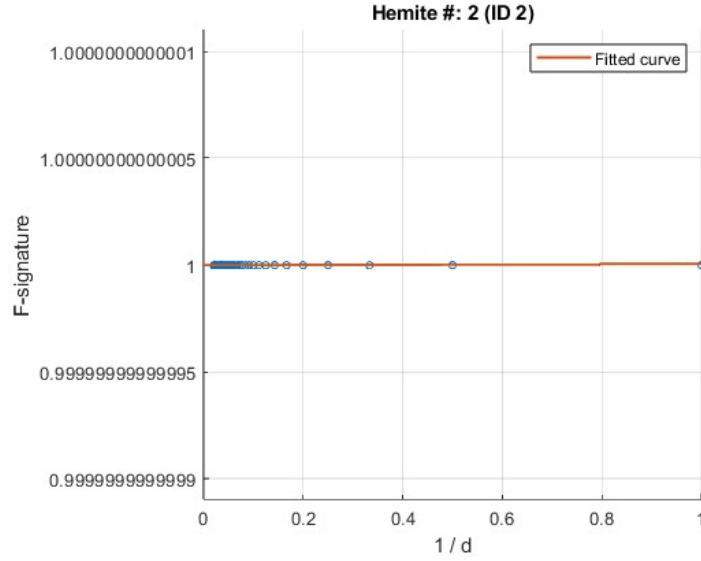


FIGURE 10. Graph of an approximation of the limit F-signature of the second Hermite polynomial

However, if the polynomial has an odd number of linear factors, then the limit F-Signature follows $\mathfrak{s}_f(t) = -x + 1$.

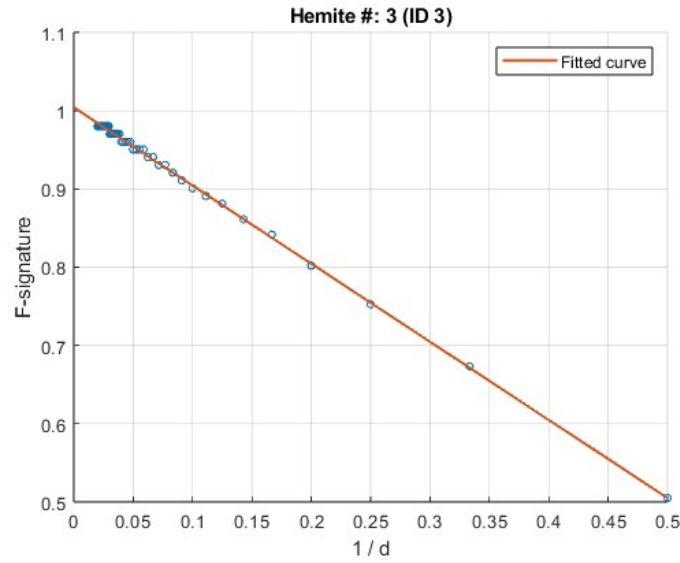


FIGURE 11. Graph of an approximation of the limit F-signature of the second Hermite polynomial

3.8. Loose Threads and Dead Ends. Our original conjecture was that the weight function of classical orthogonal polynomials was deeply intertwined in the

limit F-signature, but after discovering the black magic polynomial, we dropped it. However, it may still be a possibility. Additionally, there are many other types of orthogonal polynomials, such as Gegenbauer polynomials and Jacobi polynomials, but we did not explore these. Another idea was that the limit F-signature is related to the transverse manifold intersection numerical invariant, but we did not look into it properly. On a related note, we wanted to explore if the F-signature and limit F-signature were related to any physics-y topics. Lastly, data indicates there **could** be a polynomial to describe our products of homogenized orthogonal polynomials in three variables, but we did not explore it.

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